

Lesson 2: Classical Machine Learning as the Starting Point

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Data, prediction, and the bridge toward quantum machine learning

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Learning Outcomes

- Explain the role of data, features, target and model.
- Understand how a classical regression workflow makes a prediction.
- Recognize why data preparation is important before quantum machine learning.
- Connect the classical ML pipeline to the later quantum ML lessons.

Why a Classical ML Example Comes First

- The notebook begins with house size as a numerical feature and price as a target.
- A model is trained from examples before making a prediction.
- This gives us a familiar mental model: data $\rightarrow$ model $\rightarrow$ prediction.
- Later quantum ML lessons retain this workflow but change the feature map, circuit, or kernel.

Read the Python Code: Data and Model

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

X = np.array([500, 700, 900, 1100, 1300]).reshape(-1, 1)
y = np.array([150000, 180000, 210000, 250000, 280000])

model = LinearRegression()
model.fit(X, y)
predicted_price = model.predict([[1000]])
print(f"Predicted Price for 1000 sq ft house: ${predicted_price[0]:,.2f}")
```

- X stores house size in square feet; `reshape(-1,1)` makes it a 2D feature matrix.
- y contains target prices.
- `LinearRegression()` creates the model.
- `fit(X,y)` estimates a line from the examples.

Read the Python Code: Prediction

- `model.predict([[1000]])` asks the trained model for one new house.
- The model uses the learned relationship between area and price.
- The output is a numeric prediction, not a quantum measurement.

Interpreting the Output

Predicted Price for 1000 sq ft house: \$230,500.00

- The notebook predicts \$230,500 for a 1000 sq ft house.
- This is deterministic for the same trained model and same input.
- The prediction is an example of classical numerical computation.

From Classical ML to Quantum ML

- Classical feature: raw numbers such as age, salary or area.
- Quantum feature map: converts classical numbers into parameters of a quantum circuit.
- Quantum circuit: transforms the encoded quantum state.
- Measurement or a quantum kernel provides information back to the classical learner.

Prerequisite Bridge: What Learners Need Next

- Python variables, arrays and functions.
- Basic probability and vectors at an intuitive level.
- Meaning of training, testing and accuracy.
- Basic idea of a circuit: input $\rightarrow$ operation $\rightarrow$ output.

Beginner Activity

- Change the input from 1000 to 800, 1200 and 1500.
- Before running, predict whether price should rise or fall.
- Discuss why a model can predict reasonably even though it was not trained on exactly 1000 sq ft.

Lesson Summary

- A model learns a relationship from examples.
- Features and targets are different roles in a dataset.
- A prediction is a classical numeric result.
- Quantum ML keeps the learning workflow but introduces quantum circuits or kernels.

This classical pipeline is the reference point for the quantum lessons.

Quick Check

- ① What is the feature in this notebook?
- ② What is the target?
- ③ Why is `reshape(-1,1)` used?
- ④ What changes when we move from classical ML to quantum ML?